Exercise 8 Solution. Observe that

$$\mathbb{P}(\tau_a = n) \overset{(*)}{=} \frac{1}{2}\mathbb{P}(S_{n-1} = a-1) - \frac{1}{2}\mathbb{P}(S_{n-1} = a+1)$$

$$= \left[\frac{1}{2}\cdot\binom{n-1}{\frac{a+n-2}{2}}\cdot 2^{-(n-1)}\right] - \left[\frac{1}{2}\cdot\binom{n-1}{\frac{a+n}{2}}\cdot 2^{-(n-1)}\right]$$

$$= \frac{1}{2}\cdot 2^{-n}\cdot 2^{1}\left[\binom{n-1}{\frac{a+n-2}{2}} - \binom{n-1}{\frac{a+n}{2}}\right]$$

$(k = \frac{a+n}{2})$

$$= 2^{-n}\left[\binom{n-1}{k-1} - \binom{n-1}{k}\right]$$

$$= 2^{-n}\left[\frac{k}{n}\binom{n}{k} - \frac{n-k}{n}\binom{n}{k}\right]$$

$$= 2^{-n}\left[\frac{k-(n-k)}{n}\binom{n}{k}\right]$$

$$= 2^{-n}\left[\frac{2k-n}{n}\binom{n}{k}\right]$$

$$= 2^{-n}\left[\frac{2\left(\frac{a+n}{2}\right)-n}{n}\binom{n}{\frac{a+n}{2}}\right]$$

$$= 2^{-n}\cdot\frac{a}{n}\cdot\binom{n}{\frac{a+n}{2}}$$

$$= \frac{a}{n}\mathbb{P}(S_n = a).$$